Hydrocarbon Leak Hazard

Field Manual — Setup and Troubleshooting Reference

Use case definition. A hydrocarbon leak hazard is an unintended release of gas or liquid from seals, flanges, valves, hoses, vessels, or piping. Abnormal behavior includes detector response, visible liquid, odour, pressure loss, flow mismatch, or an increasing release rate. Troubleshooting prioritizes people, ignition control, emergency notification, remote isolation, depressurization, and verified gas-free conditions; physical repair is secondary and must follow the approved site procedure.

Manual setup and use. Use the site emergency response plan, hazardous-area classification, detector action levels, permit-to-work, gas-testing rules, and OEM/site limits. Values are training/reference values, not universal exposure or evacuation limits.

Safety note. Do not approach, tighten, open, or repair pressurized hydrocarbon equipment. Do not enter a vapour cloud. Follow alarm, muster, remote isolation, and emergency-response procedures.

Quick-use workflow

Use this sequence for every event: establish a safe operating state; capture the initial numerical readings; compare each value with the stated reference; apply the trigger-based action; correct the identified component or process path; restore operation gradually; and confirm the expected numerical result. If an OEM/site alarm or trip limit differs from a training value, the OEM/site limit governs.

Scenario 1: Mechanical seal leak

Field Condition

liquid leak 0.18 L/min; detector 12 %LEL at 3 m from pump; pressure 5.2 barg

Problem

A seal release near rotating equipment is an immediate loss-of-containment condition.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses detector above site action level or leak above 0.05 L/min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when area isolated, detector below site clearance level, leak stopped.

Expected Result

area isolated, detector below site clearance level, leak stopped

Scenario 2: Flange leak

Field Condition

visible mist; detector 28 %LEL at 5 m; upstream 42 barg; temperature 64 °C

Problem

Pressurized flange leakage creates a hazardous vapour cloud.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses detector above site alarm level.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.

3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when remote isolation and verified safe atmosphere.

Expected Result

remote isolation and verified safe atmosphere

Scenario 3: Valve stem leak

Field Condition

detector 9 %LEL at 2 m; pressure 38 barg; leak trend +2 %LEL/min

Problem

Increasing detector response near a valve stem indicates developing release.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses trend above 1 %LEL/min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when detector returns to baseline after isolation.

Expected Result

detector returns to baseline after isolation

Scenario 4: Process hose leak

Field Condition

hose spray estimated 0.7 L/min; pressure 18 barg; detector 35 %LEL at 6 m

Problem

A pressurized hose failure can escalate rapidly.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses any visible spray or detector above action level.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when remote isolation, pressure zero verified, hose replaced.

Expected Result

remote isolation, pressure zero verified, hose replaced

Scenario 5: Small liquid leak

Field Condition

drip 0.03 L/min; pooled area 0.4 m²; detector 2 %LEL; pressure 6 barg

Problem
Even a small liquid release can form vapour and spread to drains.

What Can Be Done
1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pool growth or detector above site action level.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when source isolated and no new accumulation.

Expected Result
source isolated and no new accumulation

Scenario 6: Gas leak

Field Condition
detector 42 %LEL at 4 m; pressure 31 barg; wind 2 m/s toward electrical skid

Problem
A gas cloud is moving toward an ignition source.

What Can Be Done
1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses detector above site alarm/evacuation threshold.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when remote isolation, ignition control, and clearance confirmed by authorized gas test.

Expected Result
remote isolation, ignition control, and clearance confirmed by authorized gas test

Scenario 7: Increasing gas detector reading

Field Condition
detector rises 3 to 24 %LEL in 6 min; pressure 27 barg; distance 8 m

Problem
A rapidly increasing reading indicates an active or worsening release.

What Can Be Done
1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses increase above 2 %LEL/min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when reading falling to baseline after isolation.

Expected Result

reading falling to baseline after isolation

Scenario 8: Leak during startup

Field Condition

detector 16 %LEL at 5 m; line pressure rose 0 to 12 barg in 90 s; flow 42 m^3/h

Problem

The release appears when the system is pressurized, indicating a startup leak path.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses detector above site action level.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when startup stopped, pressure isolated, detector baseline.

Expected Result

startup stopped, pressure isolated, detector baseline

Scenario 9: Leak after maintenance

Field Condition

detector 11 %LEL at flange 3 m after gasket replacement; pressure 9 barg

Problem

Post-maintenance gasket/bolting integrity is suspect.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses any confirmed detector response above site baseline.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when isolation and approved leak test passed.

Expected Result

isolation and approved leak test passed

Scenario 10: Leak near electrical equipment

Field Condition

detector 18 %LEL at 7 m; electrical panel 4 m away; pressure 22 barg

Problem

Hydrocarbon vapour is within the vicinity of an ignition-capable asset.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses detector above site action level.

2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when remote isolation and area made safe by response team.

Expected Result

remote isolation and area made safe by response team

Scenario 11: Leak near pump

Field Condition

seal area detector 14 %LEL; liquid 0.12 L/min; pump 1,480 RPM; pressure 5.4 barg

Problem

A leak at rotating equipment can contact hot surfaces or be atomized.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses leak above 0.05 L/min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pump isolated and leak repaired under approved procedure.

Expected Result

pump isolated and leak repaired under approved procedure

Scenario 12: Leak near compressor

Field Condition

detector 21 %LEL at 6 m; seal-gas pressure 1.2 barg; compressor discharge 29 barg

Problem

Low seal-gas pressure is associated with a compressor-area release.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses seal-gas differential below approved minimum.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when compressor isolated and detector baseline.

Expected Result

compressor isolated and detector baseline

Scenario 13: Hydrocarbon odour

Field Condition

odour reported 15 m from manifold; detector 4 %LEL; pressure 26 barg; wind 1.5 m/s

Problem

Odour with a detector response is treated as a possible release, not dismissed as nuisance.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses detector above site baseline.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when source isolated and gas test confirms safe condition.

Expected Result

source isolated and gas test confirms safe condition

Scenario 14: Pressure loss with suspected leak

Field Condition

header pressure falls 32 to 26 barg in 8 min; flow mismatch 18 m^3/h; detector 7 %LEL

Problem

Pressure loss and flow imbalance support a suspected leak.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure loss above 2 barg/10 min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when remote isolation and pressure stable.

Expected Result

remote isolation and pressure stable

Scenario 15: Increasing leak rate

Field Condition

liquid rate rises 0.02 to 0.16 L/min in 12 min; detector 8 to 19 %LEL; pressure 14 barg

Problem

The increasing release requires escalation and immediate isolation.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses rate increase above 0.05 L/min or detector trend above site trigger.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when release stopped and detector returns toward baseline.

Expected Result

release stopped and detector returns toward baseline